

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (CL)

December 2011

CL 207 Concrete Technology

Date: 05.12.2011, Monday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION - I

Q - 1 (a) Enlist the qualities of water to be used for making good concrete. [01]

(b) Explain the term soundness of cement and causes of same. Name the soundness test of cement. [03]

(c) State the importance of compaction of fresh concrete and enlist methods of compaction. [03]

Q - 2 (a) What is bulking of aggregate? State the effect of bulking on strength of concrete. [03]

(b) What is grading of aggregate? Explain the test performed for grading of aggregate. [04]

(c) What are admixtures? Explain the benefits of using admixture in concrete. Also explain plasticizers and its effect on properties of fresh concrete. [07]

OR

Q - 2 (a) Explain Alkali Aggregate reaction in aggregate. State factors affecting alkali aggregate reaction. [07]

(b) Explain air entraining admixture in brief. [03]

(c) What is super plasticizer? State its effect on hardened concrete. [04]

Q - 3 (a) Write a note on workability of concrete including definition, factors affecting it and methods of its measurement. [07]

(b) Explain factors affecting durability of hardened concrete. [07]

OR

Q - 3 (a) In detail, explain manufacturing process of concrete. [07]

(b) Explain the causes of sulphate attack and its methods to control the same. [03]

(c) How shrinkage affects the strength of concrete? Explain factors affecting the shrinkage. [04]

SECTION - II

Q - 4 What is mix design? Which are the properties of concrete consider in the mix design to achieve the desirable properties of concrete? [07]

Q - 5 (a) What is setting time? Explain the term initial and final setting time of Ordinary Portland Cement. [03]

(b) What is Fly Ash? Explain its cost criteria, workability and use of fly ash in mass concrete with respect to cement. [03]

(c) Explain hydration process of Cement including chemical reactions. [04]

(d) Enlist four different types of cements used in special conditions and explain any one in detail. [04]

OR

Q - 5 (a) Explain the term heat of hydration. Explain consequences it in setting process of cement. [03]

(b) From where silica fumes is obtained? Explain its influence on fresh concrete [03]

(c) Describe the role of gypsum in the hydration reaction of cement. [04]

(d) State the properties and use of Rapid Hardening Cement. [04]

Q - 6 (a) State main ingredients of concrete and their contribution in strength of concrete. Enlist five different names of special concrete. [05]

(b) Write short note on High strength concrete incorporating ingredients, properties at fresh state, properties at harden state and its importance. [05]

(c) Explain how moisture correction of moist aggregate is considered in water demand in mix design. [04]

OR

Q - 6 (a) How concrete is classified according to strength in IS: 456 – 2000? [03]

(b) Why concrete is tested on the 28th day after casting? [02]

(c) Write short note on Light weight concrete in terms of ingredients, properties at fresh state, properties at harden state and its importance. [05]

(d) In mix proportioning, it desirable to use the minimum quantity of water, why? [04]
